

1. 平衡概念

- As a system approaches equilibrium, both the forward and reverse reactions **are occurring**.
- At equilibrium, the forward and reverse reactions **are proceeding at the same rate** – **DYNAMIC !**
- Once equilibrium is achieved, the **amount** of each reactant and product **remains constant**.

Equilibrium

double arrow : $\rightleftharpoons$

2. 平衡常数

$$K_{eq} = \frac{k_f}{k_r} = \text{a constant}$$

从两端进行都可以达到

The Equilibrium Constant

- Consider the generalized reaction
 $aA + bB \rightleftharpoons dD + eE$
- The equilibrium expression for this reaction would be

$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

← products
← reactants

mol/L

- Also, since **pressure** is proportional to **concentration** for gases in a closed system, the equilibrium expression can also be written

$$K_p = \frac{(P_D)^d (P_E)^e}{(P_A)^a (P_B)^b}$$

Equilibrium

$$1 \text{ atm} = 760 \text{ torr} = 760 \text{ mm Hg}$$

- We contain $K_p = K_c (RT)^{\Delta n}$

where

$$\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$$

Equilibrium Constants and Units

➤ **Equilibrium constants are reported without units.** 没有单位

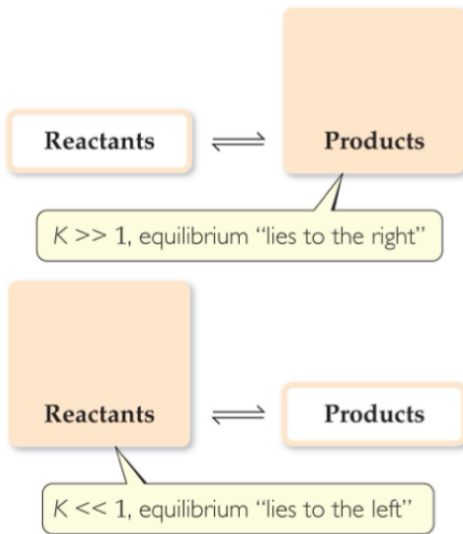
➤ Equilibrium constants is related to kinetics of a reaction as well as to the thermodynamics (Chapter 19).

➤ Equilibrium constants derived from thermodynamic measurements are defined in terms of **activities** rather than concentrations or partial pressures. 仅跟温度有关

➤ The **activity** (活度) of any substance in an *ideal* mixture is the ratio of the concentration or pressure of the substance either to a reference concentration (1 M) or to a reference pressure (1 atm). Therefore, activities have no unit but the numerical value of the activity equal to the concentration.

Equilibrium

Magnitude of K



- If $K \gg 1$, the reaction favors products; products predominate at equilibrium.
- If $K \ll 1$, the reaction favors reactants; reactants predominate at equilibrium.

以……为主

Equilibrium

① 方程反向 $K' = \frac{1}{K}$

② 方程系数加倍数 $K' = (K)^n$

③ 方程相加, $K' = K_1 \cdot K_2$

Homogeneous vs. Heterogeneous

- **Homogeneous equilibria** occur when all reactants and products are in the same phase.
- **Heterogeneous equilibria** occur when something in the equilibrium is in a different phase.

在此环境中

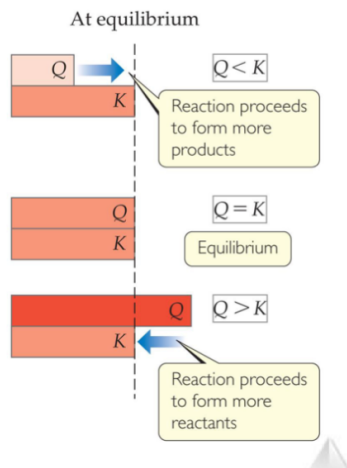
- The value used for the concentration of a pure solid or liquid substance is always 1.

⑤ 进行方向

reaction quotient 反应熵 Q

→ current condition
不一定是平衡

- Nature wants $Q = K$.
- If $Q < K$, nature will make the reaction proceed to products.
- If $Q = K$, the reaction is in equilibrium.
- If $Q > K$, nature will make the reaction proceed to reactants.



④ 勒夏特列

"If a system at equilibrium is **disturbed by a change** in temperature, pressure (volume), or the concentration of one of the components, the system will shift its equilibrium position so as to **counteract** (offset) the effect of the disturbance."

- Higher volume or lower pressure favors the side of the equation with more moles (and vice-versa).

→ 本质还是影响了浓度

- Is the reaction **endothermic** or **exothermic** as written? That matters!
- **Exothermic**: Heat acts *like* a product; adding heat drives a reaction toward reactants.

Reactants $\rightarrow$ Products + heat

- **Endothermic**: Heat acts *like* a reactant; adding heat drives a reaction toward products.

Reactants + heat $\rightarrow$ Products

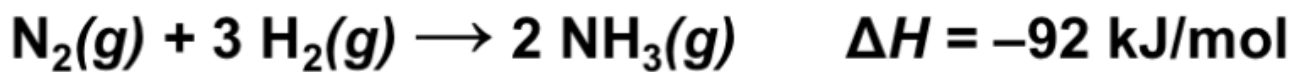
- Higher temperature $\rightarrow$ more heat $\rightarrow$ more product for endothermic reaction $\rightarrow$ higher equilibrium constant K
- Higher temperature $\rightarrow$ more heat $\rightarrow$ more reactant for exothermic reaction $\rightarrow$ lower equilibrium constant K

改变平衡的进行

且改变 K_{eq}

Equilibrium

☆ 举例 Haber Process
→ 放热

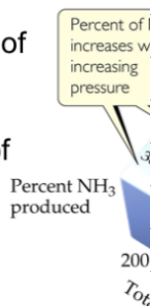


没用 T ↓ more larger K_{eq} smaller K 进行太慢

活化能很高的

The transformation of nitrogen and hydrogen into ammonia (NH_3) is of tremendous significance in agriculture, where ammonia-based fertilizers are of utmost importance.

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ammonia

还是用的液化 NH_3 移走
加快反应

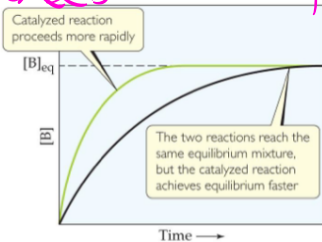
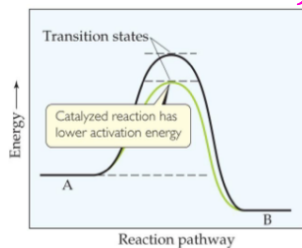
Catalysts

- **Catalysts** increase the rate of both the forward and reverse reactions.
- Equilibrium is achieved faster, but the **equilibrium composition remains unaltered.**

K 不变

未改变的

平衡组分



Equilibrium